

Predicting Pre-meal Glucose using Sparse Optimization and Time-Series features

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Meet the team

Both team members each contributed to the project, with each developing independent optimization solutions and combining both approaches together in a compare and contrast discussion of the pros and cons based on accuracy metrics. Each member discussed their approach, and reviewed each other's discussions for reasonableness and accuracy. Although we focused on two separate implementations, we have repeatedly met to discuss findings and alter our future works based on our results and progress.

Syed Ahmad Shah Implementation 1

1. Data Extraction, Transformation, and Cleaning
2. Feature Engineering from glucose, insulin, meal, exercise, and time-series event data
3. Objective Function Formulation using regularized regression models
4. Implementation of Ridge, LASSO, Elastic Net, and HistGradientBoostingRegressor
5. Time-based train/test evaluation and comparison against baseline models
6. Patient-specific modeling, residual modeling, and feature-importance interpretation
7. Visualization, result analysis, and discussion of dataset limitations and future improvements

Christopher Ognibene Implementation 2

1. Data Preparation: extracted source data and transformed into wide format, with Codes as columns, and each row indexed by Patient ID-Datetime. Filled missing values with the mean of the code per patient-day.
2. Feature engineering: Averaged four pre-meal glucose measurements
3. Target Variable: computed a 1-period forward variable, per patient.
4. Objective function: formulated a traditional mean squared error function, with L1 and L2 penalty constraints.
5. Numerical solution: programmed key optimization functions by hand with custom functions for gradient descent, L1 and L2 penalties, and elastic net objective functions.
6. Model evaluation: computed MSE, RMSE, and MAPE metrics, as well as examined residual plots and QQ-plots to assess fit.

What is our Goal?

The **goal** of the **project** is to build a **glucose prediction model** that uses the **fewest meaningful predictors** to accurately predict glucose levels keeping prediction error low. In other words, we can examine the tradeoff between accuracy of **prediction vs. sparsity** (fewest predictors). A good model should have a sufficient balance that does not overfit the training data to the extent that test error is significantly worse (e.g., worse than random guess 50%).

The value of our project is to provide alternative toolkits to medical professionals working with patients with diabetes to continuously monitor and understand expected ranges of blood sugar levels pre, and post meals and throughout the day. Accurate recording of blood glucose levels at predictable intervals throughout the day can help a patient and their health care provider achieve the best possible outcomes to managing patient health and ensuring risk of adverse events (e.g., hyperglycemia) are minimized. In addition, there is a predictive benefit to understanding the correlation of patient blood glucose levels with activity indicators in helping providers understand diabetes risk factors and estimated glucose levels for new patients.

In summary, our project seeks to add a perspective to the health community in providing a way to identify the key potential drivers behind a particular patient's glucose levels based on their recorded daily activity and inform patients to distinguish helpful from unhelpful activities to managing blood glucose levels.

Optimization Model

We explored two different implementations, described below.

Implementation 1

We predict a patient's pre-meal glucose level using prior insulin, meal, exercise, and previous glucose levels, while selecting a sparse set of predictors using a linear model:

$$y_i = \beta_0 + x_i^T \beta$$

y_i = glucose value for target observation i

x_i = feature vector for observation i

β = model coefficients

Where the shared error term between the tested Ridge, LASSO, and Elastic Net is the MSE:

$$\frac{1}{2n} \sum_{i=1}^n (y_i - \beta_0 - x_i^T \beta)^2$$

In this implementation, the model is written in penalized form rather than an explicit constrained form. However, an existing constraint, is that our model's regression coefficients cannot grow arbitrarily large, as this may lead to overfitting, additionally we encourage the model to use a smaller subset of predictors that contribute the most towards the predictions.

Implementation 2

We predict a patient's next pre-meal glucose using current insulin dosage, post-meal glucose, and exercise activity, similarly identifying the most relevant predictors that minimize out-of-sample test error.

The objective function is a least squares function, with additional L1 and L2 regularization penalty functions to increase sparsity and prioritize features with greatest influence (i.e., largest drivers behind minimizing prediction error, the difference between predicted and actual average next-period pre-meal glucose level, per patient. We are careful to avoid data leakage by having no next-period features present in our training set.

Least squares is typically sensitive to outlier data, therefore, regularization can help to create a more robust solution by identifying the most impactful features on model prediction (i.e., largest coefficients). While there are no explicit constraints in the formulation of the objective function, the constraints are the additional terms appended onto the core MSE term in the L1, L2, and Elastic Net formulas.

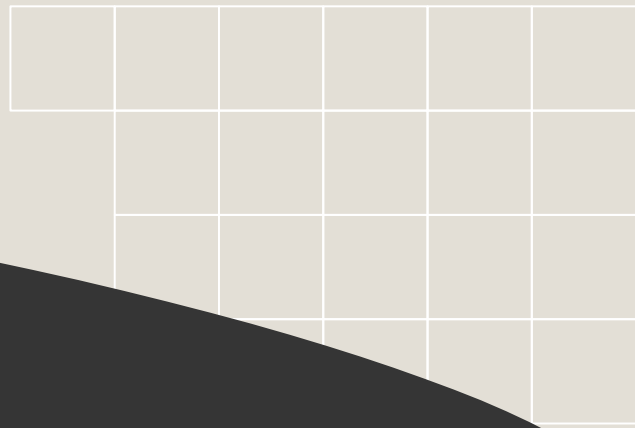
$$\text{MSE (unconstrained)} \min_w f(w) = \frac{1}{2n} \sum_{i=1}^n (x_i^T w - y_i)^2 = \frac{1}{2n} \|Xw - y\|_2^2$$

$$\text{L1 regularized: } \min_w f(w) = \frac{1}{2n} \|Xw - y\|_2^2 + \lambda \sum_{j=1}^p c_j |w_j|$$

$$\text{L2 regularized: } \min_w f(w) = \frac{1}{2n} \|Xw - y\|_2^2 + \frac{\rho}{2} \|w\|_2^2$$

$$\text{Elastic Net: } \min_w f(w) = \frac{1}{2n} \|Xw - y\|_2^2 + \lambda \sum_{j=1}^p c_j |w_j| + \frac{\rho}{2} \|w\|_2^2$$

Proposed Solution



Implementation 1

Is it convex/non-Convex

For the models implemented:

1. Ridge:
 - a. Uses a squared error term, which is convex
2. Lasso:
 - a. L1 penalty, which is convex
3. Elastic Net
 - a. L2 penalty, which is convex

As our models are convex, they won't have bad local minimas as do non-convex models, thus our solver will be more reliable.

What is the feasible set

The feasible set depends strictly on how we define to go about the problem. In the code, we use the sklearn library, which uses the penalized form: minimized prediction error + penalty.

In that way the feasible set are all possible coefficient vectors, with a penalty that discourages large or too many coefficients.

For

1. Ridge this means the square coefficient size is bounded
2. LASSO the total absolute coefficient is bounded
3. Elastic Net this means both the square coefficient size and total absolute coefficient is bounded

Why did we conclude a solution exists?

A solution exists as the objective function is continuous and regularized, and the feasible set is also convex and bounded. Additionally, the penalties prevent the coefficients from growing too large, thus the optimization problem is well constrained.

Numerical Solution using a Toolbox

From the library “scikit-learn” we utilized:

1. RidgeCV
2. LassoCV
3. ElasticNetCV
4. HistGradientBoostingRegressor (For comparison)

When implementing Ridge, sklearn solves the convex regression, and tunes the parameters through cross validation.

For LASSO and Elastic Net, sklearn utilizes coordinate descent to update one coefficient at each iteration.

Closed Form Solution ?

The models differ in regards to the closed form. As Ridge Regression is a convex quadratic function, and by taking the derivative of the objective function with respect to its coefficients and setting it to 0, we can obtain a solvable system of linear equations $\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$

However, LASSO and Elastic Net do not have closed form solution as they include an L1 penalty, which introduces a non-differentiable term when evaluated at zero.

Implementation 2

The next solution's approach employs a least-squares objective function formulation, where the goal minimizes the prediction error between the matrix product of our data and ideal model coefficients with the actual next-period pre-glucose measurement. We incorporate L1 and L2 regularization penalty functions, which are embedded constraints on the inclusion/exclusion and size of coefficients, aiming to achieve sparsity in the model (prioritize features with greatest driver for minimizing prediction error).

We consider the following:

- Prediction error is defined as the predicted average pre-glucose level (next-period) from the actual average pre-glucose (next-period) level of a patient, based on current period activity predictors.
- Reason: We will ultimately run a regularized Linear Regression algorithm, which is sensitive to outliers. Regularized least squares is robust to outliers in the underlying data, and can help identify the most impactful features in the model with the highest coefficients. However, we will also apply scaling prior to regularization to further attempt to best fit signal rather than excessive noise.
- We aim to achieve a best fit surface while accepting imperfect accuracy, in exchange for minimizing total error with a sparse (parsimonious) model with sufficient out-of-sample accuracy. (i.e., future unseen patient activity data).



Convexity

The objective function is convex. Since least squares is quadratic, we have a clear minimum, therefore, with a sufficient initial guess and step size, our algorithm is guaranteed to converge.

Why did we conclude a solution exists?

The objective function is bounded below by zero and unbounded above, therefore a global minimum exists.

Closed Form Solution ?

A solution exists as Least squares is convex, and can be solved analytically with the normal equations.

$$\hat{\beta} = (X^T X)^{-1} X^T \bar{y}.$$

However, we will use numerical solutions as we apply L1, L2, and elastic net regularization, where L1 penalty does not have a closed form solution (i.e., non-differentiable at $x=0$).

Numerical Solution using a Toolbox

For unconstrained least squares and L2 regularized least-squares, traditional gradient descent is used.

$$\beta_j \leftarrow \beta_j - \alpha \nabla g$$

For L1 regularization, we use proximal gradient descent which caps the weight update at 0 if the step is within $\alpha * \lambda$ (i.e., within vicinity of non-differentiable points ($x=0$)).

$$\text{prox}_{\lambda}(\beta_i) = \begin{cases} \beta_i - \lambda & \text{if } \beta_i > \lambda \\ \beta_i + \lambda & \text{if } \beta_i < -\lambda \\ 0 & \text{otherwise} \end{cases}$$

Validating Solution

About the Dataset:

We acquired the dataset from the **UC Irvine Machine Learning Repository** at the link here, <https://archive.ics.uci.edu/dataset/34/diabetes>. This was an open-source dataset provided for initial use for academic researchers and medical professionals, and made public to students or practitioners of machine learning for academic or personal project use. **The data contains details on basic physiology and patho-physiology of diabetes mellitus and its treatment. The dataset is composed of 70 diabetes patient records collected from electronic and paper records. Patients are indexed by 1-70.** Each patient has several weeks' to months' worth of outpatient care data.

The feature names are:

Data-Codes ('Code'): a listing of the codes used in the data sets.

Date in MM-DD-YYYY format: i.e., date of recording

Time in XX:YY format: i.e., time of recording

Value: i.e., blood glucose level

Codes will be the main set of features we will use in our optimization problem approaches, as our goal is to predict glucose levels with respect to the key codes.

Several key features are defined below:

33 = Regular insulin dose - short-acting insulin (works soon after injection, often around meals)

34 = NPH insulin dose - intermediate-acting insulin (lasts longer than regular, and is used for background insulin plan)

35 = UltraLente insulin dose - Long-acting insulin (Different role than regular insulin, as its not for immediate spikes)

48, 57 = Unspecified blood glucose measurement - Glucose was measured, but not tied to meal context

Our Solution:

For both solutions, the following was the process followed to calibrate each of our objective functions to minimize prediction error for glucose level.

- **Data Extraction:** We extracted the original source data from the compressed zip file into individual CSV files (one per patient) using Python's data compression and analysis libraries. We aggregated the individual CSV files into one data set.
- **Data Transformation:** data is converted from 'long' to 'wide' format into the more traditional data set: the Code column was expanded into individual columns per code, corresponding to one recording (i.e., patient-timestamp).
- **Define the target:** Use rows with glucose codes 58, 60, 62, 64 (Pre-glucose measurements), as this is our objective.
 - **Implementation 1:** Utilizes historical features
 - **Implementation 2:** average of pre-glucose measurement codes per timestamp, then forward-shift by patient to create a forward-looking target. Fill remaining N/A values with the mode. (i.e., most occurring measurement)
- **Build features:** Use only events that happened before each target measurement.
- **Clean the data:** Handle malformed dates, irregular times, and nonnumeric glucose values if needed.
- **Split by time:** Train on earlier observations and test on later observations.
- **Compare models:**
 - **Implementation 1:** candidate models Mean predictor, Ridge regression, LASSO, Elastic net, PCA baseline
 - **Implementation 2:** candidate models Unconstrained LS, L1 regularized, L2 regularized, and Elastic Net
- **Evaluate:** MSE, RMSE, Number of selected predictors, tradeoff between sparsity and accuracy

Implementation 1

Data Processing and Feature Engineering:

- The data was separated by **patient** and included the **date, time, code, and value**.
- To better fit the data for a regression model, the date was **normalized** to be “**Days since start**”, and Time to “**MinuteOfDay**” as the model would be better able to relate to continuous data as opposed to utilizing OneHotEncoding.
- Prior activity, insulin, and exercise indicators were engineered (shown right), which were used to train and calibrate the optimization models with prior activity as predictors of current pre-glucose levels.
- For each patient:
 - If Code = Target Glucose Measurement, form a unique observation by Code, Minute
 - Engineered features spanned prior 6, 8, and 24 hours from current pre-glucose measurement (i.e., do not equal current time to avoid data leakage)
- Total of 10,346 rows out of 58,661 raw rows.

Feature	Lookback Window	Type
typical_meal_6h	number of prior "66" events over last 6 hours	Diet
more_meal_6h	number of prior "67" events over last 6 hours	Diet
less_meal_6h	number of prior "68" events over last 6 hours	Diet
regular_insulin_8h	Sum of Measurements of "33" over last 8 hours	Dosage
nph_insulin_24h	sum of measurements of "34" over last 24 hours	Dosage
ultralente_insulin_24h	sum of measurement of "35" over last 24 hours	Dosage
typical_exercise_6h	number of prior "69" events over last 6 hours	Exercise
more_exercise_6h	number of prior "70" events over last 6 hours	Exercise
less_exercise_6h	number of prior "71" events over last 6 hours	Exercise
prev_glucose	Most recent pre or post glucose measurement	Glucose Level
hours_since_prev_glucose	Number of hours elapsed since last pre or post glucose measurement	Glucose Level
hypo_symptoms_24h	number of past "65" events over last 24 hours	Glucose Level
glucose	current level of pre-glucose [the target]	Glucose Level
special_event_24h	number of past "72" events over last 24 hours	Unspecified

Implementation 1

- **Target definition:** We created one-hot encoded target variables for each of our four pre-glucose measurement features, because these will be our decision variables (1/0), with our actual target glucose level as stored in the 'Value' field.
- **Train/Test split:** we use the first 80% of rows stratified by patient for training, and the remainder for testing
- **Baseline 1: simple mean predictor** - Predict glucose using the average training glucose. This gives a weak baseline to beat (a "naive" prediction)
- **Baseline 2: previous glucose predictor** - Predict the next pre-meal glucose using the most recent previous glucose.
- **Ridge regression** - Fit ridge regression on all engineered features. Purpose: closed-form mathematical baseline, handles correlated predictors, good comparison point.
- **LASSO regression** - Fit LASSO on the same feature set. Purpose: sparse feature selection, convex optimization model, easier interpretation.
- **Elastic net regression** - Fit elastic net on the same feature set. Purpose: combines LASSO sparsity with ridge stability usually better when predictors are correlated likely your best main model.
- 5-fold cross validation was used, along with null value imputation using the median, and standard scaling (Z-score)

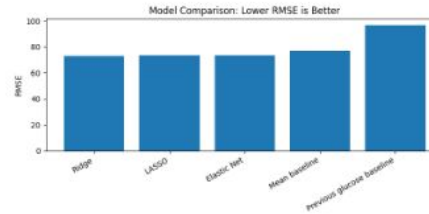


Figure: Comparison of Model Performance

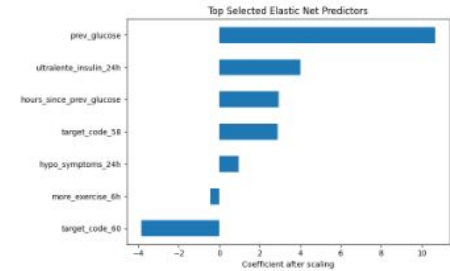


Figure: Feature Importance

	Model	MSE	RMSE	MAE	Selected Predictors
2	Ridge	5332.484106	73.023860	58.338130	21.0
3	LASSO	5397.191415	73.465580	59.063308	7.0
4	Elastic Net	5413.498230	73.576479	59.236969	7.0
0	Mean baseline	5892.664434	76.763692	62.582119	NaN
1	Previous glucose baseline	9377.593643	96.837976	71.344254	NaN

Implementation 1

- We conducted the same analysis as described on the prior slide, however, with the key difference of including **patient id as a feature** (i.e., one-hot encoded patient id as columns). Therefore, our decision variables include not just the 4 pre-glucose binary variables, but also **70 binary patient variables**.
- **Train/Test split:** rather than splitting by patient id strata, we split by patient based on the DAYS feature, with the first 80% of recordings for training, the remainder for testing.
- The five models are fit as described on the prior slide.

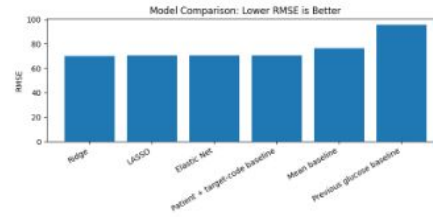


Figure: Comparison of Model Performance
(with Patient ID and Code as Predictors)

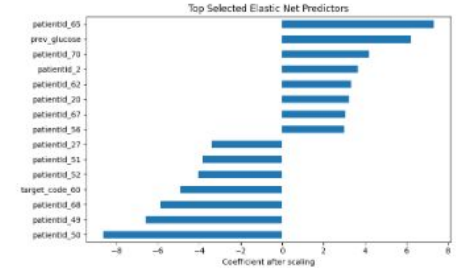


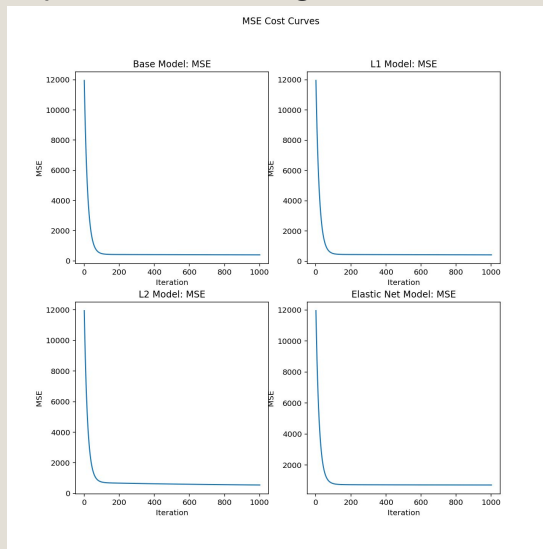
Figure: Feature Importance

	Model	MSE	RMSE	MAE	Selected Predictors
3	Ridge	4924.237908	70.172914	55.358211	86.0
4	LASSO	4952.979558	70.377408	55.741738	43.0
5	Elastic Net	4952.979558	70.377408	55.741738	43.0
2	Patient + target-code baseline	4954.792381	70.390286	54.691903	NaN
0	Mean baseline	5891.123437	76.753654	62.505170	NaN
1	Previous glucose baseline	9194.173502	95.886253	70.567823	NaN

Implementation 2

- **Baseline 1:** unconstrained least squares. This solution is the least parsimonious as all of our features are included in prediction.
- **LASSO (L1) regression** - least-squares objective function plus the L1 norm (sum of absolute values of coefficients), which shrinks coefficients to 0. Can result in more parsimonious models.
- **Ridge (L2) regression** - least-squares plus the L2 norm (sum of squares of coefficients), which shrinks coefficients but not necessarily to 0. Considered a middle-state.
- **Elastic net (L1 and L2) regression** - taken to incorporate both L1 and L2 norm terms to reduce and potentially eliminate features from the model.

Learning Curves: as we can see, all methods converge with respect to a max iteration of 1000. Examining the performance metrics provides further insight to the differing precision of the algorithms.



	MSE	RMSE	MAPE
Base Model (Train)	803.308006	28.342689	0.155339
Base Model (Test)	796.774938	28.227202	0.154794
L1 Model (Train)	805.108091	28.374427	0.155374
L1 Model (Test)	798.691875	28.261137	0.154808
L2 Model (Train)	823.709199	28.700334	0.153718
L2 Model (Test)	817.221248	28.587082	0.152514
Elastic Net Model (Train)	858.745954	29.304367	0.153863
Elastic Net Model (Test)	853.847314	29.220666	0.153147

Key Insights



Implementation 1 Discussion

What are the key observations you made from the solutions as applied to the problem?

- In the general model (no patient ID relation) the Elastic Net model selected 7 predictors, whilst in the patient-ID version selected 43 predictors.
- If features are engineered from prior exercise, diet, and glucose levels and patient id is incorporated as well, accuracy is ~6% more accurate (i.e., elastic net performance, 59.23 MAE) as compared to without patient id feature (55.74 MAE).
- A one-size-fits all model is not appropriate for this problem, as different individuals with different body compositions, metabolisms, and reactions to injections can severely alter expected results.

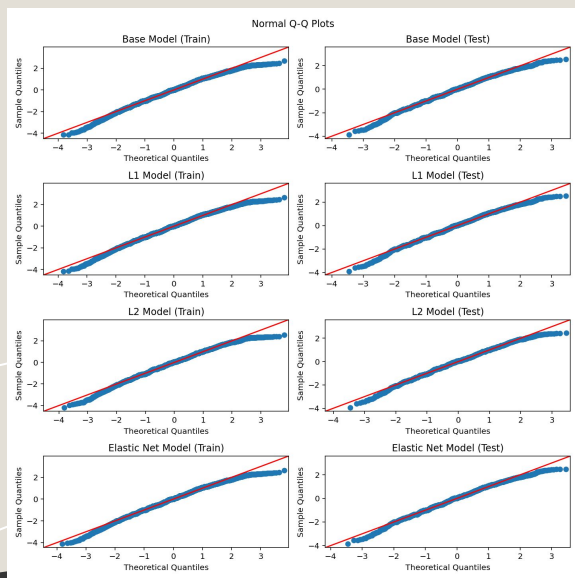
What does your solution mean for the problem?

- Our implementation shows that the pre-meal glucose predictions can indeed be framed as sparse regression problem, where the number of predictors can be kept small enough to retain accuracy.
- Pre-meal glucose isn't random: it contains predictive structure from previous observations (ie: insulin timing and overlapping injections)
- A linear optimization model with the current available data is not able to severely over perform simple baselines, but may be able to create sampling data for new patients.
- However, the model incorporating patient id as a feature may be valuable more so for existing patients when you're seeking to track patient health over time, rather than for new/prospective patients when data is unavailable or very limited.

Implementation 2 Discussion

What does your solution mean for the problem?

- The models with the **lowest overall error (MAPE)** is the **L2 regularized model**, with 15.25% test error, and while all coefficients were retained, only 9 of the 19 (excluding bias term) had values greater than 1, indicating greater importance. However, all models can be seen as reasonable based on similarity in test accuracy.
- This implementation approach illustrates the tradeoff for simplicity with more involved feature engineering (as shown in Implementation 1), while achieving potentially sufficient accuracy for given a practitioner's use case.

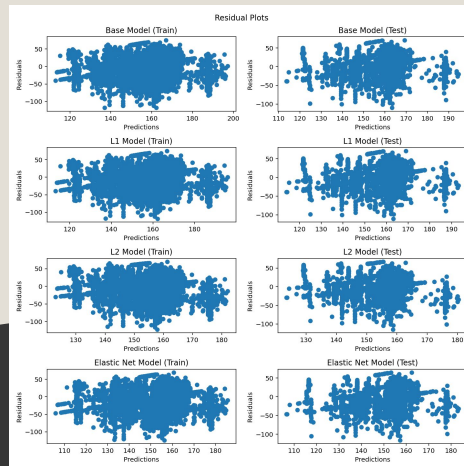


What are the key observations you made from the solutions as applied to the problem?

- From the residual plots below for train and test data, we note generally good fit (i.e., largely appears to be random pattern), however, there may be potential non-linearity in our features (slight potential uptrend in residuals), but overall, this does not appear to be of large concern.
- The close normality in residuals (seen from the Normal Q-Q plots on the left), assures us of sufficiently good linear regression fit.

Any Limitations/When can these models be used?

- A more simplistic approach if a patient does not have extensive health history (e.g., new/prospective patient). We could also consider transformations to our features to try to adjust for non-linearity).
- Can be used as a rough baseline to compare with more advanced predictive models or feature engineering approaches.



Major Conclusions

Was your approach viable?

- The exercise of exploiting two meaningfully different optimization approaches was valuable as either approach may be sufficient based on the desired level of sophistication for the modeler / health administration.
- A simpler, less feature engineered approach may be desired either for simple baselining of patients or for new/prospects, whereas a more nuanced model can be better served to established patient tracking over time.

Did you run into any difficulties such as infeasibility, numerical instability, unexpected results, etc.

- Data ingestion was a key deliberation in our process, as we did not have traditional time series (i.e., gaps).
 - As a result, the simpler implementation, while objectively lower error, may likely incorporate more bias than the more complex feature-engineering approach, trading off for increased variance.
- 

Suggestions for Improvement and Future Work

If you had more time and resources what would you have done to solve the problem?

- The most important limiting factor in our implementation is our dataset. Medical information is highly protected and restrictive, and our current chosen dataset is one of a few publicly available datasets that includes labelled patients. To better predict glucose levels, we would require the following features:
 - Meal carbohydrate amount
 - Patient specific physiology (ie: age, gender, weight, metabolism, etc.)
 - Medical Background
 - Exercise intensity
 - Sleep and Stress
 - Meal context
 - Body composition
- If provided additional time, we would be able to communicate with a point-of-contact from the organization “[Synapse](#)”, which curates a list of medical and patient information for research, which contains more information about patient specific information. This would provide our models with more diverse and complex data from which deeper relationships and insights could be built from.
- We would also explore a baseline time series model (ie., autoregressive) or a Long-Short Term Memory Neural Network for time series forecasting. Given that true patient data likely tends to incorporate seasonality/cyclical, these alternative models can potentially fit signal better than traditional least squares regression.